

Development and evaluation of FINEX off-gas capture and utilization processes for sustainable steelmaking industry

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ABSTRACT

FINEX process was proposed as an environmentally friendly option for steelmaking. To further its environmental friendliness, this study proposes and evaluates three options for FINEX off-gas (FOG) capture and utilization: post-combustion CO₂ capture (Case 1), pre-combustion CO₂ capture (Case 2), and methanol production following the pre-combustion CO₂ capture (Case 3). These options are designed through simulation and are comparatively evaluated in terms of their net CO₂ emissions and economics by conducting CO₂ lifecycle assessment (LCA) and techno-economic analysis (TEA). In the latter, CO₂ avoidance cost is estimated to assess the economic feasibility of each option. Case 1 was observed to give the largest amount of CO₂ emission reduction, which is 35% lower than the current FOG utilization method of burning it for electricity generation. Case 2 showed the lowest CO₂ avoidance cost of 68 USD/ton. Although Case 3 was estimated to give higher net CO₂ emissions than Case 1 and higher CO₂ avoidance cost than Case 2, sensitivity analysis revealed that the outcome depends strong on the global warming impact of the mixed electricity, carbon credit price, and methanol price and it can become an attractive option in the future.

1. Introduction

Steelmaking industry currently faces a serious challenge brought by several global trends such as price increases of raw materials and stricter environmental regulations, especially for emissions like CO₂, NO_x, and SO_x (POSCO E&C, Primetals Technologies Austria GmbH, 2020). As the traditional blast furnace (BF) route of producing steel falls short of meeting such a challenge, the smelting reduction technologies have been developed as an alternative ironmaking route (Cavaliere, 2019). One of the commercially deployed smelting reduction technologies is the FINEX process by POSCO, South Korea. The advantages of the FINEX process over the BF process are well documented (Yi et al., 2019): it circumvents the need for the sintering and coking processes thereby requiring lower investment costs. In addition, low-grade raw materials (coal fines and iron ore fines) can be used, reducing raw material costs. Furthermore, the FINEX process releases only 9% of NO_x, 6% of SO_x, and 59% of dust, compared to the conventional BF route (POSCO E&C, Primetals Technologies Austria GmbH, 2020; Yi et al., 2019).

However, the FINEX process is not known to give lower CO₂

emissions than the conventional BF route. As the industry is being pressured to dramatically lower its greenhouse gas emission due to climate change, capture and utilization of the off-gases may be needed to ensure its sustainability. For the conventional BF route, many technologies have been proposed to lower the carbon emission, the most common being off-gas valorization and integration with carbon capture and storage (CCS) technologies. In off-gas valorization, off-gases from the BF process are converted into various valuable products after being properly cleaned (Deerberg et al., 2018; Deng and Adams II, 2018, 2020; Gao et al., 2020; Kim and Han, 2020). In addition, several CO₂ capture methods have been investigated within the confines of an integrated steel plant. Arasto et al. investigated the use of post-combustion capture with amine scrubbing to capture CO₂ from the off-gas stream of the BF process and the results showed a significant CO₂ reduction potential (Arasto et al., 2013). Similarly, Nordic CCS Competence centre reported that a huge amount of carbon dioxide emission can be avoided by two different pre-combustion capture methods of MEA absorption and the Selexol process in iron and steel production (Onarheim et al., 2015). Compared to the off-gas streams from the BF process, which are the blast

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furnace gas (BFG) and the coke oven gas (COG), the FINEX off-gas (FOG) can be advantageous in that it has a lower proportion of nitrogen and a higher proportion of CO₂, allowing for an easier capture of CO₂ (Carpenter, 2012). Furthermore, FOG contains higher proportions of CO and H₂ than BFG, making it more suitable for producing chemicals from the off-gas. However, few studies have investigated the FOG off-gas utilization from the viewpoint of reducing the CO₂ emission so far.

In this study, various CO₂ emission reduction methods are applied to FOG: post-combustion capture, pre-combustion capture, and pre-combustion capture followed by methanol production. After they are simulated using Aspen plus software, their CO₂ emissions are estimated through CO₂ lifecycle assessment (LCA) and unit cost of production is calculated through techno-economic analysis (TEA) by considering both capital cost and operating cost. Finally, CO₂ avoidance cost is obtained for each option from the LCA and TEA results. These results are compared not only with each other, but also with the current practice of burning the off-gas for electricity generation. This helps to assess which of CO₂ mitigation methods can reduce CO₂ the most or in the most economical way. To sum up, the objectives of the study are as follows.

- (1) To propose three options for CO₂ mitigation for the FINEX process by combining various CO₂ capture and utilization methods.
- (2) To compare the CO₂ emissions of the proposed options quantitatively through CO₂ LCA and to evaluate their economic feasibilities based on the CO₂ avoidance cost estimated from TEA.

2. FINEX process

The FINEX process, one of commercial smelting reduction technologies, directly uses iron ore fines and non-coking coal, hence it circumvents the need for raw material treatment steps like sintering and coking processes (Yi et al., 2019). As illustrated in Fig. 1, fine ore is pre-reduced in a multi-stage fluidized-bed reactor and then charged to a melter-gasifier, where coal is injected to supply energy for melting iron and to produce reducing agent (CO) through gasification. The reducing agent is then supplied to the fluidized-bed reactors for pig iron extraction. FINEX off-gas, FOG, is generated from the multi-stage fluidized-bed reactor. Hot metal produced from the melter-gasifier passes through a basic oxide furnace to remove impurities, then is cooled and solidified to form a steel slab.

Typical characteristics of FOG are compared with other steelmaking off-gases in Table 1. Conditions of these off-gases are mostly provided by POSCO, but temperature, pressure, and chemical composition of COG are from the literature (Deng and Adams, 2018). Production rate of off-gases refers the amount of gases currently transported to the power plant, which is what remains after using off-gases as a heat energy source for the sub-processes of the steelmaking process. The amount of sulfur is a critical index, because an additional desulfurization process is required for excess sulfur that exceeds the standard, before a power plant or a reactor. FINEX process emits less sulfur compound than BF process because most of the sulfur is captured in the slag, and sulfur compounds are washed out in the fluidized-bed reactors (Yi et al., 2019).

3. Process development

At current time, off-gases from a steelmaking process are mostly used to generate electricity through the combined cycle power plant (CCPP), for a supply back to the steelmaking process, as shown in Fig. 2. Since the produced electricity is generally insufficient to produce a corresponding amount of steel slab, electricity from an external source is supplemented. We assume that FINEX process produces two million tons

Table 1

Typical characteristics of steelmaking off-gases (Deng and Adams II, 2018).

Component	FOG	BFG	COG
Temperature (°C)	40	40	35
Pressure (bar)	1.16	1.16	1.44
Production rate (Nm ³ /ton-Slab)	1035	733	23
LHV (kcal/Nm ³)	1360	801	4317
Chemical composition (volume fraction)			
%CO	28.4	24.4	8.4
%CO ₂	40.3	24.8	3.1
%H ₂	11.7	2.2	56.4
%N ₂	12.9	48.5	2.3
%CH ₄	1.5	–	26.6
%O ₂	–	–	0.3
%C ₂ H ₂	–	–	2.9
%H ₂ O	5.2	–	–
H ₂ S (ppmv)	22	–	3780
CS ₂ (ppmv)	–	–	87
C ₄ H ₄ S (ppmv)	–	–	30

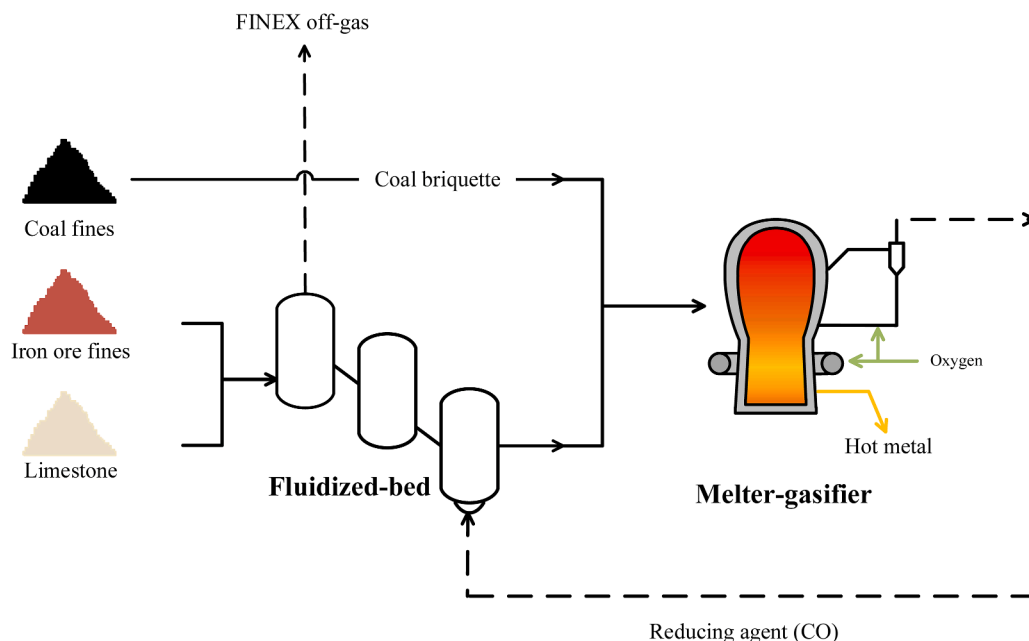


Fig. 1. FINEX route for steel production (POSCO E&C, Primetals Technologies Austria GmbH, 2020).

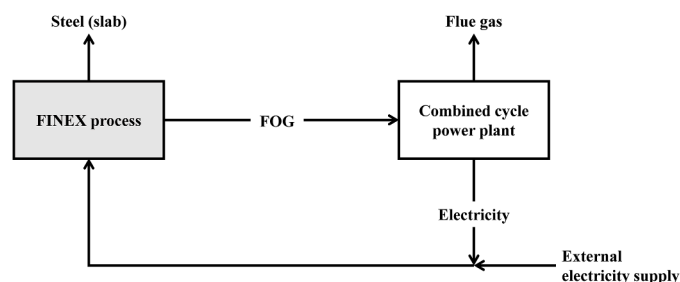


Fig. 2. Configuration of the current representative FOG utilization.

of steel slab per year in this work. For sustainable FINEX process, three potential FOG utilization methods are suggested and examined in this study, which are represented in Fig. 3: post-combustion capture, pre-combustion capture, and methanol synthesis following pre-combustion capture. All three utilization processes involve CO₂ capture, and captured CO₂ is assumed to be sequestered deep underground. Note that the FINEX process itself is not affected by the additional off-gas capture and utilization processes, because the same amount of electricity is supplied to the FINEX process in all three cases, for fair comparisons. Major assumptions for the suggested processes are summarized in Table 2.

Case 1 incorporates post-combustion CO₂ capture into the current FOG utilization method, because significant amounts of carbon dioxide are emitted through the flue gas from the combined cycle power plant. Generally, post-combustion flue gas is at atmospheric pressure and has a lower concentration of CO₂. As it has a low driving force for CO₂ separation, chemical absorption, e.g., amine scrubbing, is the standard. The amine-based absorption of CO₂ is a proven technology and has been commercialized on a large scale (Rao and Rubin, 2002). In the designed process, Monoethanolamine (MEA) is used to remove CO₂ from the flue gas. Captured CO₂ is assumed to be sequestered deep underground so it goes through a conditioning to meet the conditions for CO₂ transport and storage.

Alternatively, CO₂ can be captured from FOG before it is combusted in the CCPP, which is referred to as pre-combustion capture. Case 2 is the integration of the FINEX process with the pre-combustion CO₂ capture. When separating carbon dioxide directly from FOG, physical absorption using a solvent like dimethyl ethers of polyethylene glycol (DEPG) is a preferable option over the MEA absorption given the higher driving force for the separation; the CO₂ partial pressure is much higher as its concentration in the FOG is much higher than in the flue gas. Physical absorption in general consumes much less energy in the regeneration step compared to the MEA absorption (Jansen et al., 2015). Furthermore, the FOG stream contains more than 20 ppm of hydrogen sulfide (H₂S) which presents a problem for the MEA absorption. The designed process involves Selexol process to physically absorb the acid gas under high pressure (53.5 bar in this study). The treated FOG is then used to generate electricity through the CCPP, while the separated CO₂ stream goes through the CO₂ conditioning step for pipe transportation. Since the pressure of treated FOG is high because of the high operating pressure of the Selexol process, it generates more electricity than the untreated FOG. Thus, more electricity is generated than needed by the FINEX process and the excess is assumed to be sold.

Given that FOG contains significant H₂ and CO, and less nitrogen and sulfur than BFG and COG respectively, potentials are there to convert the gas into valuable chemicals after the CO₂ removal. Case 3 involves a methanol production following the pre-combustion capture from FOG. The reason for choosing methanol as a target product among various possible products is that its market size is large, which is expected to reach 43 billion US dollars in 2027, and it can be an intermediate to other products such as olefin, DME, and acetic acid (Report Linker, 2022). The water-gas-shift (WGS) reactor is introduced to increase the H₂/CO ratio to 2–3, which is suitable for the methanol synthesis. The

amount of CO₂ in FOG increases as the result of WGS reaction. Excess CO₂ and H₂S are removed by the Selexol process to improve the conversion and to prevent catalyst deactivation. Additionally, since the purge stream from the methanol reactor still contains energy intensive gases, it is burnt to generate electricity for the FINEX process.

4. Process model

All three options proposed in this study include some of the following sub-processes: combined cycle power plant, MEA absorption, Selexol process, methanol synthesis, and CO₂ conditioning process. These sub-processes are simulated through the process simulation software Aspen Plus, and a detailed description of the process models is in Supplementary materials.

Combined cycle power plant generates electricity by gas turbine and steam turbine systems. The configuration and assumptions of CCPP used in this study are from the literature (Chase and Kehoe, 2000; Field and Brasington, 2011). The operating pressure of the gas turbine is assumed to be 16 bar and the discharge pressures of the three steam turbines are 29 bar, 4.5 bar, and 0.07 bar, respectively (Field and Brasington, 2011).

In Case 1, the amine-based absorption process captures CO₂ from the flue gas. 30 wt.% aqueous solution of monoethanolamine (MEA) is used as an absorbent and 90% of CO₂ is captured. For the simulation of the MEA absorption process, a rate-based model is used with liquid film discretization (Chung and Lee, 2020). The Electrolyte Nonrandom Two-liquid (ENRTL) model is used for thermodynamic properties (Chung and Lee, 2020). The ENRTL model is the thermodynamic equilibrium model that calculates activity coefficients based on the excess Gibbs free energy and is suitable for electrolyte systems such as amine scrubbing processes (Bollas et al., 2008). Reactions and kinetic parameters related to CO₂ absorption/desorption by MEA are described in Supplementary materials (Aspen Plus, 2008).

Selexol process was named after its solvent DEPG, whose commercial name is Selexol. DEPG physically absorbs acid gas under high pressure. To model the Selexol process, we use the Perturbed-Chain Statistical Associating Fluid (PC-SAFT) thermodynamic property method and the PCSKIJ binary interaction parameters from the literature (Field and Brasington, 2011). The PC-SAFT predicts the non-ideal behavior of various mixtures based on perturbation theory. The PCSKIJ parameters are in Table S.3. The two-stage Selexol process is used to absorb H₂S and CO₂ separately (Ahn et al., 2014; Field and Brasington, 2011). It is operated under 53.5 bar and 90% of CO₂ is separated from FOG.

In Case 3, methanol is produced from treated FOG. It is synthesized at 260 °C and 52 bar, and two distillation columns are then used to purify methanol (Guil-Lopez et al., 2019). Two reactions are considered for methanol synthesis: water-gas-shift reaction and CO₂ hydrogenation (Bussche and Froment, 1996; Mignard and Pritchard, 2008; Simoes Van-Dal and Bouallou, 2013). Chemical equations of the reactions are as below:



Parameters related to the kinetics of methanol synthesis are described in Table S.4 (Simoes Van-Dal and Bouallou, 2013).

Captured carbon dioxide needs a conditioning process to meet the quality specifications for CO₂ transport. Pipeline is the most efficient way of CO₂ transport, and the supercritical state of CO₂ is preferred when it is transported via pipelines (Leung et al., 2014). The quality specifications for CO₂ transport are shown in Table 3 (Glass, 2019; Hu, 2011). To meet the conditions, the captured CO₂ stream is dehydrated and compressed to 150 bar (compressed to 90 bar using compressor and a pump is used to increase the pressure of liquid CO₂ to 150 bar).

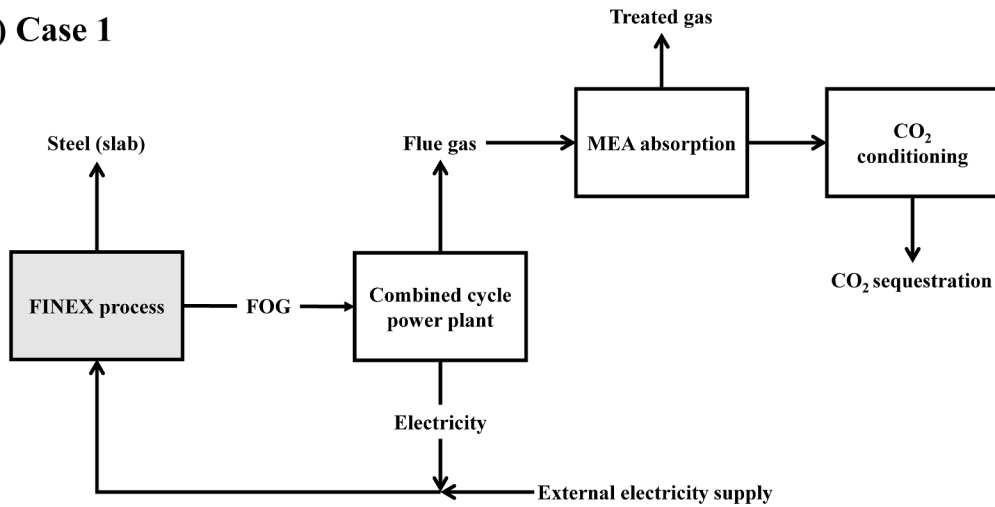
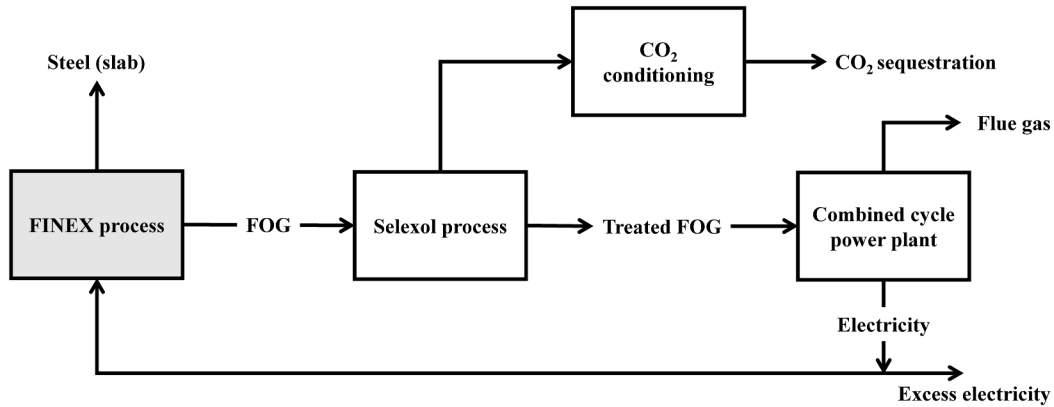
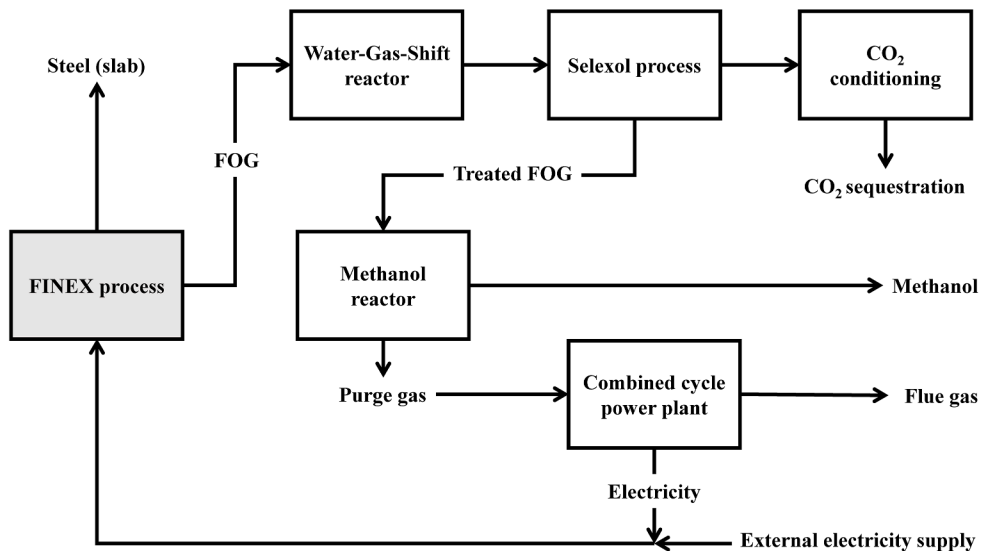
(a) Case 1**(b) Case 2****(c) Case 3**

Fig. 3. Configurations of three suggested FOG utilization processes for sustainable FINEX process.

5. Results and discussion

5.1. Simulation results

To compare the proposed processes on an equal basis, one ton of steel

slab is set as a functional unit, and the steelmaking process integrated with each CO₂ mitigation process is identified as the system boundary. Detailed simulation results of the FOG utilization processes are in Table 4, but raw materials of the FINEX process (iron ores, coal, etc.) are neglected. Electricity requirement for the FINEX process, except the

Table 2

Major assumptions for three suggested FOG utilization processes.

Assumptions
- 245 tons of steel slab is produced per hour (About two million tons per year).
- All CO ₂ capture processes involved are operated at 90% of CO ₂ removal from their inlet streams.
- Captured CO ₂ is sequestered deep underground.
- The same amount of electricity (0.70 Gcal/ton-Slab) is supplied to the FINEX process in all cases.

Table 3CO₂ quality specifications for transport.

Specifications	
Product	At least 98 mol.% of CO ₂
Water (Glass, 2019)	No more than 50 ppm by volume
Hydrogen Sulfide (Glass, 2019)	No more than 20 ppm by volume
Temperature (Glass, 2019)	Not exceed 49 °C
Typical CO ₂ transport pressure in pipe (Hu, 2011)	150 bar

amount supplied by Linz-Donawitz converter gas (LDG), is 0.70 Gcal/ton-Slab. The amount of generated electricity from the combined cycle power plant in the current FOG utilization method is 0.68 Gcal/ton-Slab, corresponding to 97% of the electricity requirement.

Case 1 generates the identical amount of electricity as the current FOG utilization method, and it captures 1.35 ton of CO₂/ton-Slab from the flue gas through the amine (MEA) absorption process. Since the amine absorption process consumes a large amount of steam to regenerate the amine solution, Case 1 consumes 1.19 Gcal of steam per ton-Slab.

Case 2 is estimated to generate 0.75 Gcal of electricity through CCGP, which is higher than the current method due to the high-pressure stream from the Selexol process. The excess electricity is considered as a by-product. This option captures 0.70 ton of CO₂/ton-Slab, which is far less than the previous option. It is because this system captures CO₂ prior

Table 4

Simulation results of various FOG utilization processes.

Component	Sub-process	Current method	Case 1	Case 2	Case 3
Raw materials (ton/ton-Slab)	Air	3.88	3.88	4.94	1.11
	Process water	0.01	1.14	0.01	0.41
	MEA	–	0.00	–	–
	DEPG	–	–	0.00	0.00
Energy consumption (Gcal/ton-Slab)	TEG	–	0.00	0.00	0.00
	Steam	CCGP	0.00	0.00	0.00
		CO ₂ capture (MEA or Selexol)	–	1.19	0.05
		CO ₂ cond.	–	0.00	0.00
		WGS	–	–	0.00
	Cooling water	Methanol	–	–	0.09
		CCGP	0.54	0.54	0.47
		CO ₂ capture (MEA or Selexol)	–	0.97	0.20
		CO ₂ cond.	–	0.18	0.07
	Coal	WGS	–	–	0.21
		Methanol	–	–	0.44
		CCGP	0.00	0.00	0.03
		CO ₂ capture (MEA or Selexol)	–	0.00	0.00
	Electricity	CO ₂ cond.	–	0.00	0.00
		WGS	–	–	0.12
		Methanol	–	–	0.19
		CCGP	0.00	0.00	0.00
		CO ₂ capture (MEA or Selexol)	–	0.02	0.18
		CO ₂ cond.	–	0.10	0.04
		WGS	–	–	0.00
		Methanol	–	–	0.00
By-product	Electricity requirement for FINEX process	0.70	0.70	0.70	0.70
	Electricity generation from CCGP	–0.68	–0.68	–0.75	–0.20
	Methanol (ton/ton-Slab)	–	–	–	0.17
CO ₂ capture (ton/ton-Slab)		–	1.35	0.70	1.04

to power generation when a significant portion of the carbon in FOG is not in the form of CO₂. The system consumes less steam and instead consumes 0.22 Gcal of electricity per one ton-Slab, which is mainly for compressing FOG stream up to 53.5 bar before the Selexol process.

Lastly, Case 3 captures 1.04 ton of CO₂ after increasing the amount of CO₂ through WGS reactor. It generates only 29% of the required electricity because it produces 0.17 ton of methanol/ton-Slab instead. Therefore, this option consumes the highest amount of electricity of the three proposed systems, mostly due to the additional electricity supply needed for the FINEX process.

5.2. CO₂ lifecycle assessment result

In this study, a functional unit of CO₂ LCA is 1 ton of steel slab. To analyze CO₂ emissions of the FOG utilization processes with a ‘cradle to gate’ system boundary, CO₂ emissions derived from the preparation of raw materials (FOG) and the FINEX process itself are also considered. They are derived from mass and energy balances (not covered in this study for confidentiality) of the steelmaking plant of POSCO. In terms of indirect CO₂ emissions, which are induced by the consumption of electricity, coal, LP steam, etc., global warming impact (GWI) factors are used. Table 5 lists the GWIs of various energy sources and raw materials. Since this study is based on the FINEX process commercialized in South

Table 5

Global warming impact of various components assumed in this study.

Types	Component	GW
	Electricity ^a	0.48 ton CO ₂ -eq./Gcal
	Coal ^b	0.38 ton CO ₂ -eq./Gcal
	LP steam ^b	0.26 ton CO ₂ -eq./Gcal
	Cooling water	Neglected
	Process water	Neglected
	MEA ^c	3.83 ton CO ₂ -eq./ton

^a (Climate Transparency, 2021).

^b (US EPA, 2022).

^c ecoinvent Version 3.

Korea, the GWI of electricity is assumed to be that of a mixed electricity of South Korea at current time (Climate Transparency, 2021). The detailed methodology of CO₂ LCA is described in Supplementary materials.

CO₂ LCA results of the current FOG utilization and the three proposed options for lowering the CO₂ emission are shown along with the contribution distribution of their components in Fig. 4 and detailed results are summarized in Table 6. The current FOG utilization method emits 2.78 ton of CO₂ per ton of steel slab produced, and direct emissions account for more than 80%. All three cases show reduction in direct emissions by capturing CO₂, while indirect emissions increase by the utility consumptions of the additional processes. Among the three options, Case 1 shows the largest net CO₂ emission reduction, 0.98 ton CO₂-eq./ton-Slab. It is because Case 1 captures CO₂ after most of carbon in FOG is converted to the form of CO₂. In Case 3, direct emission is further reduced by converting some of the carbon in FOG to methanol. Moreover, reduction effect (credit for producing products replacing conventionally made products with CO₂ footprints) is present for the two options, Case 2 producing excess electricity and Case 3 producing methanol. The GWI of conventional methanol production is assumed as 0.76 ton CO₂-eq./ton to calculate the CO₂ reduction effect of methanol (Pérez-Fortes et al., 2016). Net CO₂ emission reduction is feasible for all three proposed cases, because the decrease in direct emissions are greater than the increase in indirect emissions.

5.3. Techno-economic analysis results and CO₂ avoidance cost

In practice, it is difficult to commercialize CO₂ mitigation methods without economic merits despite potential CO₂ emission reduction. Therefore, it is important to find the most cost-effective CO₂ emission reduction method among various available options. In this section, economic feasibilities of the suggested FOG utilization processes are analyzed and compared to each other to identify which system reduces CO₂ emissions at the lowest cost. We estimate the operating expenditure (OPEX) and capital expenditure (CAPEX) using material and energy balances obtained from process modeling and simulation. The functional unit chosen is 1 ton of steel slab, and system boundary is the FINEX process integrated with the various CO₂ capture and utilization methods, as in the lifecycle assessment. However, it is difficult to divulge the production cost of steel at POSCO for the sake of confidentiality. Thus, the techno-economic analysis (TEA) of this study focuses on the cost of off-gas utilization, excluding the cost of steel production itself. Table 7 summarizes the unit costs of utilities; the costs of electricity, LP steam, and process water are assumed as those of South Korea, considering the location of the FINEX steel plant. Other basic assumptions and parameters used in the TEA are summarized in Supplementary

materials.

Table 8 represents the detailed techno-economic analysis results for the FOG utilization methods. Annualized investment cost is calculated from the equipment costs, and variable operating costs are derived from the consumed amounts of raw materials and utilities. For the current method, annualized investment cost is neglected by assuming that the steel company already owns the facilities related to the CCPP. Additional unit cost of off-gas utilization signifies the difference between the unit cost of each case and the unit cost of the current method. Additional revenue is the profit obtained from selling by-products such as electricity and methanol. CO₂ avoidance cost is obtained by subtracting additional revenue from additional unit cost and dividing it by the amount of CO₂ emission reduction (Roussanaly, 2019):

$$\text{CO}_2 \text{ avoidance cost} = \frac{\text{Additional unit cost} - \text{Additional revenue}}{\text{Amount of CO}_2 \text{ emissions avoided}} \quad (3)$$

$$\begin{aligned} \text{Additional unit cost} \\ = \text{Unit cost of each case} - \text{Unit cost of the current method.} \end{aligned} \quad (4)$$

Although CO₂ emissions decreased the most in Case 1, the CO₂ avoidance cost is the highest at 139 USD per ton of CO₂ avoided. Variable operating cost of energy consumption accounts for 69% of the unit cost of the process, and the MEA absorption step occupies 89% due to its high heat energy demand for solvent regeneration. Case 2 provides only 0.60 ton of CO₂ reduction per ton of steel, but it has the lowest CO₂ avoidance cost of 68.4 USD/ton CO₂-eq. This is because Case 2 captures CO₂ directly from FOG, which has a high proportion of CO₂, not mixed with nitrogen, thus resulting in much less energy consumption for the CO₂ capture. In addition, the process generates excess electricity using the high-pressure gas from the Selexol process. The revenue of excess electricity is calculated as 4.59 USD/ton-Slab, contributing further reduction in CO₂ avoidance cost.

Additional unit cost of Case 3 is 140 USD/ton-Slab, which is larger than the other two cases. Since FOG produces methanol instead of electricity, the FINEX process requires 0.49 Gcal/ton-Slab of more electricity to produce steel slab, accounting for 50% of the variable operating cost. In addition, the process consumes a large amount of electricity, especially in the Selexol process and CO₂ conditioning. Nonetheless, Case 3 results in a lower CO₂ avoidance cost than Case 1 due to the revenue by selling methanol. The revenue of methanol is highly dependent on the methanol price. The value in Table 8 is based on 392 USD/ton-methanol, the average price over five years (June 2017 ~ May 2022) (Mathenex, 2022). As a result, the CO₂ avoidance cost of this option is about 86 USD/ton CO₂-eq.

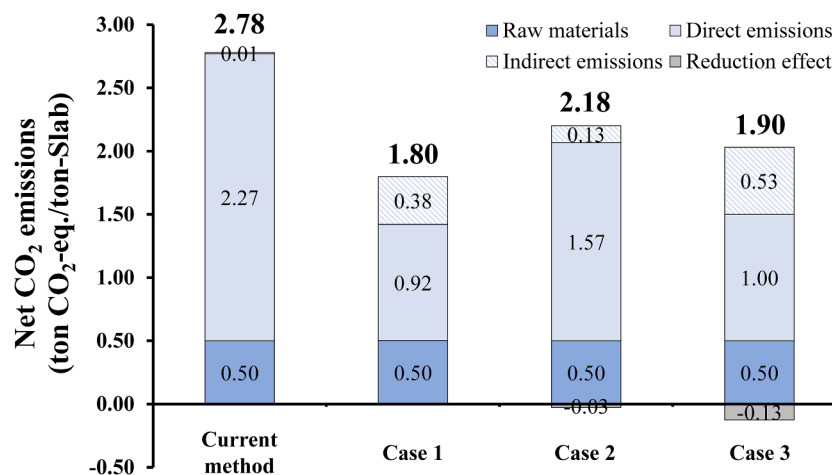


Fig. 4. CO₂ LCA results comparison of FOG utilization processes.

Table 6Detailed CO₂ LCA results of FOG utilization processes (Unit: ton CO₂-eq./ton-Slab).

Type		Sub-process	Current method	Case1	Case 2	Case 3
Raw material		FINEX process	0.50	0.50	0.50	0.50
		CCPP	0.00	0.00	0.00	0.00
		CO ₂ capture (MEA or Selexol)	–	0.00	0.00	0.00
		CO ₂ cond.	–	0.00	0.00	0.00
		WGS	–	–	–	0.00
		Methanol	–	–	–	0.00
Direct emissions		FINEX process	0.77	0.77	0.77	0.77
		CCPP	1.50	0.00	0.72	0.12
		CO ₂ capture (MEA or Selexol)	–	0.15	0.08	0.11
		CO ₂ cond.	–	0.00	0.00	0.00
		WGS	–	–	–	0.00
		Methanol	–	–	–	0.00
Indirect emissions	Steam	FINEX process	0.00	0.00	0.00	0.00
		CCPP	0.00	0.00	0.01	0.00
		CO ₂ capture (MEA or Selexol)	–	0.31	0.01	0.02
		CO ₂ cond.	–	0.00	0.00	0.00
		WGS	–	–	–	0.00
		Methanol	–	–	–	0.02
	Electricity	FINEX process	0.01	0.01	0.00	0.24
		CCPP	0.00	0.00	0.00	0.00
		CO ₂ capture (MEA or Selexol)	–	0.01	0.09	0.11
		CO ₂ cond.	–	0.05	0.02	0.03
		WGS	–	–	–	0.00
		Methanol	–	–	–	0.00
	Coal	FINEX process	0.00	0.00	0.00	0.00
		CCPP	0.00	0.00	0.01	0.00
		CO ₂ capture (MEA or Selexol)	–	0.00	0.00	0.00
		CO ₂ cond.	–	0.00	0.00	0.00
		WGS	–	–	–	0.04
		Methanol	–	–	–	0.07
Reduction effect			–	–	–0.03 (Electricity)	–0.13 (Methanol)
Total			2.78	1.80	2.18	1.90

Table 7

Unit cost of utilities assumed in this study.

Utility type	Unit cost
Electricity (Global Petrol Prices, 2021)	82 USD/Gcal
Coal (Trading Economics, 2020)	12 USD/Gcal
LP steam (Yeongnam Energy Service, 2022)	70 USD/Gcal
Cooling water (Chung et al., 2018)	1.50 USD/Gcal
Process water*	0.32 USD/ton

* Provided by POSCO, 2020.

5.4. Sensitivity analysis on external variables

The results of this study are affected by several factors such as global warming impact (GWI) of mixed electricity, carbon credit price, and methanol price. For instance, as a large amount of electricity is consumed, the amount of CO₂ emission reduction varies greatly depending on how clean electricity is used. Since all three cases reduce CO₂ emissions of steelmaking process, their economic feasibilities improve considering the benefit associated with carbon credit. Besides, the methanol price has a strong effect on the economic feasibility of Case 3. The base case uses 0.48 ton CO₂-eq./Gcal for the electricity GWI, which is the current value of South Korea's electricity mixture. Methanol price is assumed as 392 USD/ton and carbon credit price is not considered for the base case. Sensitivity analysis results for GWI of mixed electricity, carbon credit price, and methanol price are discussed in the rest of the section.

5.4.1. GWI of mixed electricity

GWI of mixed electricity depends on the combination of various fuels used to generate electricity (Table 9). China has a high electricity GWI, because China produces most of its electricity from coal, while UK has a very low electricity GWI due to the high proportions of nuclear energy and renewable energy for its electricity generation (Climate

Table 8

Detailed TEA results of FOG utilization processes.

Type	Sub-process	Current method	Case 1	Case 2	Case 3
Unit cost of annualized investment cost (USD/ton-Slab)		Neglected	25.7	14.6	33.9
Unit cost of variable operating cost: Raw materials (USD/ton-Slab)		0.00	1.16	0.00	0.13
Unit cost of variable operating cost: FINEX process (electricity) (USD/ton-Slab)	FINEX process (electricity)	1.46	1.46	0.00	40.3
Unit cost of variable operating cost: CCPP (USD/ton-Slab)	CCPP	0.81	0.81	2.63	0.00
Unit cost of variable operating cost: CO ₂ capture (MEA or Selexol) (USD/ton-Slab)	CO ₂ capture (MEA or Selexol)	–	85.7	18.4	23.4
Unit cost of variable operating cost: CO ₂ cond. (USD/ton-Slab)	CO ₂ cond.	–	8.25	3.47	4.83
Unit cost of variable operating cost: WGS (USD/ton-Slab)	WGS	–	–	–	1.97
Unit cost of variable operating cost: Methanol (USD/ton-Slab)	Methanol	–	–	–	9.65
Unit cost of fixed operating cost (USD/ton-Slab)		1.08	16.5	10.2	29.5
Unit cost of each case (USD/ton-Slab)		3.36	140	49.3	144
Additional unit cost of off-gas utilization (USD/ton-Slab)		–	136	46.0	140
Additional revenue (USD/ton-Slab)		–	–	4.59	64.9*
CO ₂ emission reduction (ton CO ₂ -eq./ton-Slab)		–	0.98	0.60	0.88
CO ₂ avoidance cost (USD/ton CO ₂ -eq.)		–	139	68.4	86.2

* Methanol price: 392 USD/ton.

Transparency, 2021). As shown in Fig. 5, the amount of CO₂ emission reduction is inversely proportional to the electricity GWI. While the capture-only cases (Case 1 and Case 2) show only 4 ~ 7% increases in the reduction effect compared to the base case when the UK's GWI is used, Case 3 shows a 24% increase and overtakes Case 1 in terms of the total reduction. This is because power consumption accounts for 71% of the large indirect emissions of the Case 3.

Table 9Changes in CO₂ emission reduction according to electricity GWI.

Electricity GWI (Climate Transparency, 2021) (ton CO ₂ -eq./Gcal)		Case 1	Case 2	Case 3
UK	0.20	1.01	0.65	1.09
EU	0.25	1.01	0.64	1.05
US	0.40	0.99	0.62	0.94
Korea	0.48	0.98	0.60	0.88
China	0.62	0.97	0.58	0.77

5.4.2. Carbon credit price

As the carbon credit price is expected to increase gradually according to the global demand for reducing CO₂ emissions, we examine how carbon credit price influences the economic feasibility of each case. A carbon credit price range of up to 150 USD per metric ton of CO₂ was considered, based on the values found in the annual energy outlook report released by U.S. Energy Information Administration (EIA, 2022). The highest carbon fee that the report estimated is 130.67 USD/ton-CO₂ in 2050 assuming 5% annual increases starting from 35 USD estimated for 2023. Fig. 6 shows how the gross revenue of each case increases with increasing carbon credit price. The gross revenue is calculated as in Eqs. (5) and (6). It is assumed that the reduced amount of carbon dioxide emission is directly linked to the carbon credit revenue.

Gross revenue

= Additional revenue

− Additional unit cost + Carbon credit revenue

(5)

Carbon credit revenue

= Carbon credit price × CO₂ emission reduction

(6)

When no carbon credit revenue is considered, all three cases require investment rather than making a profit. However, as carbon credit price increases, the gross revenue of all cases rises and turns a profit at some point. CO₂ avoidance cost defined in Eq. (3) is exactly at this point. Case 1 makes the least gross revenue (mostly the greatest loss) in this credit price range despite its highest CO₂ emission reduction. This is because the carbon credit revenue is not enough to compensate for its high operating costs until the credit price reaches 139 USD/ton-CO₂. Both Case 2 and Case 3 show more viable results, each turning in earning profits at the credit price of 68.4 and 86.2 USD/ton-CO₂, respectively. When the carbon credit price crosses 126 USD/ton-CO₂, Case 3 starts to make greater gross revenue than Case 2 due to its greater CO₂ emission reduction per one ton of steel slab.

5.4.3. Methanol price

In Section 5.4.2, we identified that Case 3 is the most profitable option on the condition that carbon credit price is high. This can be further enhanced by high methanol price. As shown in Fig. 7, high

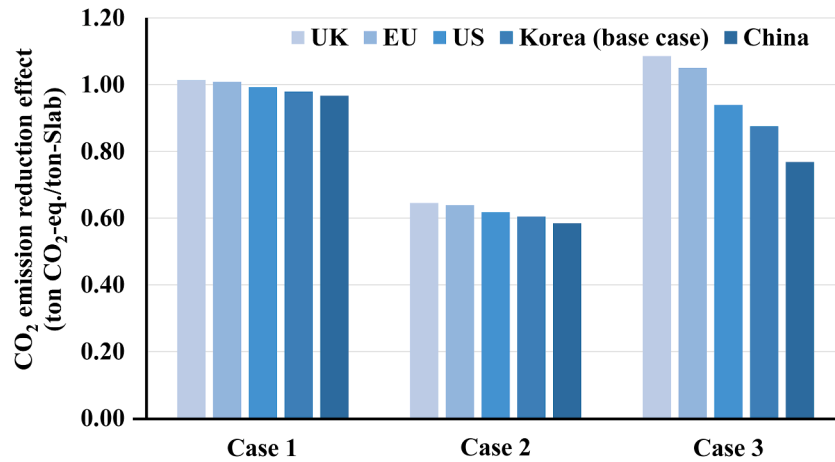
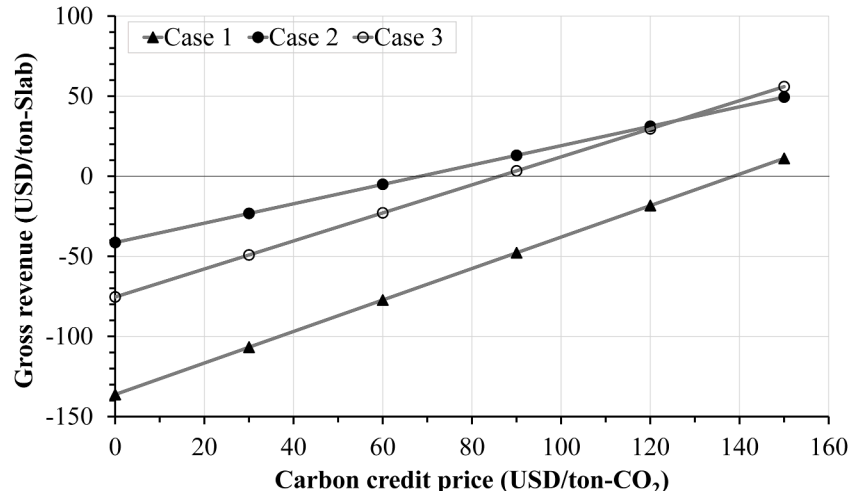
Fig. 5. CO₂ emission reduction effects of three mitigation options depending on regions.

Fig. 6. Effects of carbon credit price on gross revenue of three FOG utilization processes.

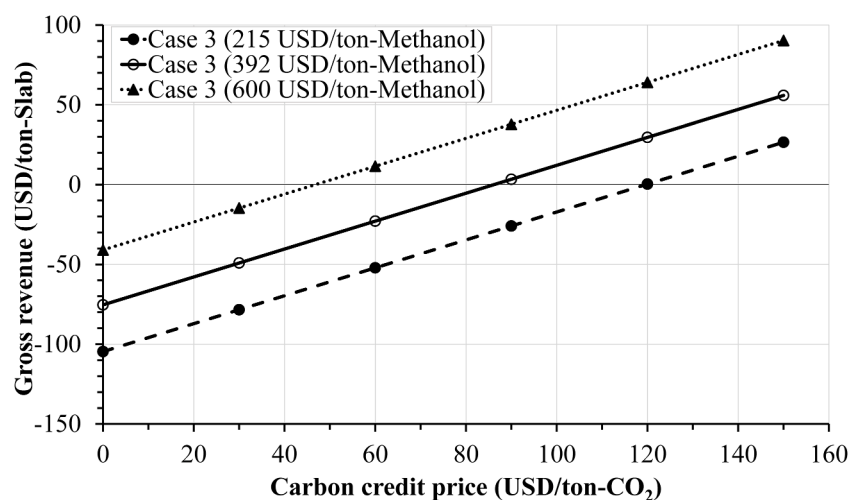


Fig. 7. Effects of methanol price on gross revenue of Case 3.

Table 10

Changes in CO₂ avoidance cost of Case 3 according to methanol price.

Methanol price (USD/ton)	CO ₂ avoidance cost (USD/ton CO ₂ -eq.)
215	120
392	86
600	47

methanol price significantly improves the economic feasibility of the methanol production option (Case 3). The price of methanol has varied between 215 USD/ton and 600 USD/ton over the past five years, with an average of 392 USD/ton (Mathenex, 2022). Table 10 shows the changes in the CO₂ avoidance cost of Case 3 with changing methanol price, which can be derived from Fig. 7. When the methanol price is at its high value of 600 USD/ton, the cost of avoiding 1 ton of CO₂ drops to a value as low as 47 USD. If the CO₂ reduction effect is increased to 1.09 ton CO₂-eq./ton-slab with the UK's electricity GWI, the CO₂ avoidance cost of Case 3 with methanol price 600 USD/ton is further reduced to 38 USD/ton, which is the lowest among the three CO₂ mitigation methods suggested in this study.

6. Conclusion

In this study, we examined the possibility of a sustainable steel-making industry through the utilization of FINEX off-gas (FOG) which has not been widely discussed in the literature. This study proposes three alternative methods of handling FOG to cope with the increasing CO₂ reduction target of the iron and steel industry. CO₂ lifecycle assessment and techno-economic analysis are conducted to estimate and evaluate the CO₂ emission reduction effect and the avoidance cost of each method.

All three proposed methods lowered the carbon dioxide emission compared to the current practice of burning the off-gas for electricity generation, even after accounting for all the indirect emissions. Among the options, Case 1 (post-combustion capture) showed the largest CO₂ mitigation. The CO₂ LCA result of the case is 1.80 ton CO₂-eq./ton-Slab, which is just 65% of the current practice. Case 2 (pre-combustion capture) and Case 3 (methanol production) showed 22% and 31% less CO₂ emissions, respectively. The sensitivity analysis for global warming impact (GWI) of mixed electricity revealed that the CO₂ mitigation effect of the Case 3 is significantly influenced by the electricity GWI. In regions such as UK, whose electricity GWI is only 42% of Korea's, the CO₂ reduction effect of the case surpasses the reduction effect of Case 1 by 7%.

Although additional costs are inevitable when CO₂ capture or off-gas

valorization is integrated into FINEX process, it is expected that these additional processes will help the economics of the entire steelmaking process amid the wave of rising carbon tax or credit. Case 2 is the most economical means to lower the CO₂ emission at current time, with the lowest CO₂ avoidance cost among three options (68 USD/ton CO₂-eq.). The reason is that the process uses less energy for capturing CO₂ and generates extra profit by selling excess electricity. It is also the least costly option per ton of steel produced when no carbon credit is considered. Though the CO₂ avoidance cost of Case 3 is higher than Case 2 at current time, Case 3 remains as a promising option for FOG utilization in the future according to the sensitivity study which revealed that the gross revenue (per ton of steel slab produced) of the Case 3 could be the largest among the three options, under the scenario of high carbon credit price or high methanol price.

CRediT authorship contribution statement

Woohyun Jeong: Conceptualization, Methodology, Writing – original draft. **Jaeseo Lee:** Conceptualization, Writing – review & editing. **Chang-Kuk Ko:** Formal analysis, Investigation. **Sang-Ho Yi:** Formal analysis, Investigation. **Jay H. Lee:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Supplementary materials

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